Thesis

Algorithms and Iterative Methods for Infinite Games

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2025

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Acknowledgments

This thesis is based on papers written with Tomáš Kroupa and Sara Vannucci. I would like to thank the organizers of the Barrande fellowship programme for the opportunity to study at LAAS-CNRS, and to Didier Henrion for his guidance during my short research stay.

Contents

1 Introduction

This thesis covers our published results on the solutions of games with infinite action spaces. It focuses on a generically applicable algorithm, and shows how the powerful underlying techniques extend to applications in robotics. Finally, it outlines the future direction for our research.

In practice, the inherent complexity of game-theoretical concepts often necessitates simplifying assumptions to model interactions between agents. There exist several classes of games with known theoretical results, allowing us to map our problem to one of those. Interesting interactions in the real world care little about game theory, however, and they appear arbitrarily far from any of the convenient classes of games. In scenarios where such reductions would be too drastic, we may still gain some insight into players' behavior with ad hoc heuristics, such as by discretizing the action spaces. While these approaches can be useful, they may lack robustness and guarantees.

In this thesis, we explore methods capable of providing guarantees or certificates on the quality of their solutions across a broad range of games. Additionally, we explore methods for global optimization, which appears as a very common sub-task in game theory. The ability to check global optimality is, for example, useful in answering whether an agent in an interaction is behaving rationally.

Finding global optima of functions over a set in the absence of any assumptions is a hard problem; even approximating such solutions is hard. Nevertheless, there exist classes of problems where global optimization is feasible. We demonstrate how Lasserre's hierarchies and the Spatial Branch and Bound method can be effectively applied to solve tasks in game theory and robotics. Our research indicates that the practicality of global methods remains largely unexplored despite the potential.

2 Strategies for Global Optimization

The analysis of competitive interactions between agents often leads to challenging mathematical optimization problems [3]. Game theory traditionally assumes that agents are motivated solely by the goal of minimizing convex cost functions [1]. This assumption is primarily a mathematical convenience to ensure the tractability of optimization problems and the existence of well-known solution concepts. When this assumption is violated, computing various stable states of interactions becomes difficult, and even determining their existence can be computationally hard [4]. Similarly, when agents can choose from an infinite number of distinct actions, finding actions that result in favorable outcomes reduces to global optimization over nonconvex functions.

Nevertheless, there are combinations of reasonable solution concepts and classes of games that are both tractable and widely applicable. Allowing agents to play randomized strategies guarantees the existence of stable points but still poses the challenge of global optimization. In this chapter, we introduce two prominent approaches to global optimization: Lasserre's hierarchies and the spatial Branch and Bound method. In later chapters, these methods will enable us to solve games with infinite action spaces and nonconvex cost functions. Furthermore, we will demonstrate their utility in the context of robotics.

2.1 Lasserre Hierarchies

The moment-SOS hierarchies are a basis of methods for constrained polynomial optimisation. More specifically, the methods will be able to find the certifiably global minimum of a polynomial over a compact set defined by polynomial inequalities. In some instances, they can even recover some minimisers. All of this is achievable under some mild assumptions on the structure of the feasible set, such as having a bound on the norm of the feasible elements. Under such constraints, each level of the hierarchy corresponds to a dual[1] pair of semidefinite programs which provide a monotonically nondecreasing sequence of lower bounds that generically converge [12] to the optimum. The current downsides of the hierarchies are that: the bounds on the convergence are not yet well understood; semidefinite solvers are not as well optimised for this task; and the recovery of minimisers of sufficient precision may be practically challenging.

A basic semialgebraic set $S(g_1, \dots, g_k) = \{\mathbf{x} \in \mathbb{R}^n \mid g_i(\mathbf{x}) \geq 0 \ \forall i = 1, \dots, k\}$ is a finite intersection of superlevel sets defined by polynomials g_i for some positive integers k and n. We further assume that the set has the Archimedean property, meaning it is certifiably compact (e.g., one of the $g_i(\mathbf{x})$ takes the form $a - \|\mathbf{x}\|^2$ for some positive a).

[1]Strong duality holds if the original constraint set has an interior point, or if either program is strictly feasible.

Let f be a polynomial and Z be a basic semialgebraic set. We are interested in finding the global minimum of the following optimization problem:

$$\min_{\mathbf{x}\in Z} \quad f(\mathbf{x}).$$

Two complementary techniques form the basis of polynomial optimization methods called the Moment-SOS hierarchies [7]. The first approach, which will later form the inner-approximation, is to maximize a lower-bound α on the condition that $f - \alpha$ is non-negative on a semialgebraic set Z:

$$\begin{aligned} \max_{\alpha\in\mathbb{R}} \quad & \alpha \\ \text{s.t.} \quad & f(x) - \alpha \geq 0 \quad \forall x \in Z. \end{aligned}$$

The future outer-approximations will be found while attempting to minimize the Lebesgue integral of f:

$$\min_{\mu\in\mathcal{P}(Z)} \quad \int_Z f \, \mathrm{d}\mu.$$

where $\mathcal{P}(Z)$ is the set of regular Borel probability measures on Z. The two presented programs are convex reformulations of the original problem, and their optimal values agree exactly. It has been proven by Riesz [10] that a truncated sequence of h moments represents integration with respect to a measure supported on a compact set K, exactly when it produces nonnegative values for every polynomial of degree at most r which is nonnegative on K. The practical versions of these problems are semidefinite programs which are their strenghtenings and relaxations respectively.

2.1.1 Existence of Representing Measures

To manipulate measures algorithmically, we will represent them as linear functionals. Every linear functional l can be encoded by a sequence of moments $y_\alpha = l(x^\alpha)$. The space of linear functionals is larger than the space of regular Borel measures; however, we will need a systematic way to carve out just those linear functionals that represent integration with respect to a measure. This is known as the K-moment problem.

The integral of a polynomial with respect to a measure can be written as a function on the moments of the measure. With a multi-index $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_m) \in \mathbb{N}^m$ we use the standard notation $\mathbf{x}^{\boldsymbol{\alpha}} = x_1^{\alpha_1} \cdots x_m^{\alpha_m}$ to define monomials. The $\boldsymbol{\alpha}$-th *moment* of a measure $\mu \in \mathcal{P}(S)$ is the real number

$$\mu'(\boldsymbol{\alpha}) = \int_S \mathbf{x}^{\boldsymbol{\alpha}} \, \mathrm{d}\mu.$$

The infinite sequence $\boldsymbol{\mu}' = (\mu'(\boldsymbol{\alpha}))_{\boldsymbol{\alpha}\in\mathbb{N}^m}$ is the *moment sequence* and μ is called a *representing measure*. The moment problem is the characterization of those real sequences $\mathbf{y} = (y_{\boldsymbol{\alpha}})_{\boldsymbol{\alpha}\in\mathbb{N}^m}$ which are moment sequences, that is, $\mathbf{y} = \boldsymbol{\mu}'$ for some μ. This problem can be decided by checking the positivity of moment expressions via *moment* and *localizing* matrices.

For any $d \in \mathbb{N}$, we define the set of multi-indices truncated to degree d as

$$\mathbb{N}_d^m = \{\boldsymbol{\alpha} \in \mathbb{N}^m \mid \alpha_1 + \cdots + \alpha_m \leq d\}.$$

Let $\mathbf{y} = (y_\alpha)_{\alpha\in\mathbb{N}^m}$ be a real sequence and $d \in \mathbb{N}$, the *moment matrix of order d* is the matrix $M_d(\mathbf{y})$ with rows indexed by $\boldsymbol{\alpha} \in \mathbb{N}^m_d$ and columns indexed by $\boldsymbol{\beta} \in \mathbb{N}^m_d$ such that

$$(M_d(\mathbf{y}))_{\alpha\beta} = y_{\alpha+\beta}.$$

The polynomials g_i defining the semialgebraic set $S(g_1, \dots, g_k)$ can be written as

$$g_i(\mathbf{x}) = \sum_\gamma c_\gamma \mathbf{x}^\gamma$$

for some real coefficients c_γ, where only finitely many of them are nonzero. Then, the *localizing matrix of order d* is the matrix $M_d(g_i, \mathbf{y})$ with rows indexed by $\boldsymbol{\alpha} \in \mathbb{N}^m_d$ and columns indexed by $\boldsymbol{\beta} \in \mathbb{N}^m_d$, where

$$(M_d(g_i, \mathbf{y}))_{\alpha\beta} = \sum_\gamma c_\gamma y_{\alpha+\beta+\gamma}.$$

When a real symmetric matrix A is positive semidefinite, we write $A \succeq 0$. Moment sequences are characterized by positive semidefinitness of moment and localizing matrices [7, Theorem 2.44(b)]. Specifically, a real sequence $\mathbf{y} = (y_\alpha)_{\alpha\in\mathbb{N}^m}$ with $y_\mathbf{0} = 1$ is a moment sequence with a Borel representing measure supported in $S(g_1, \dots, g_k)$ if, and only if,

$$\begin{aligned} M_d(\mathbf{y}) \succeq 0 \quad & \forall d \in \mathbb{N} \\ M_d(g_k, \mathbf{y}) \succeq 0 \quad & \forall d \in \mathbb{N}, \forall k \in K. \end{aligned} \tag{2.1}$$

2.1.2 Nonnegative Polynomials Using Sums of Squares

On the dual side, we represent polynomials by their coefficients. As before, we need sufficient conditions on the coefficients so that we can algorithmically decide the question of nonnegativity on a set. To do this, we will use preorders: just as ideals preserve the set of zeros, preorders preserve positivity on sets.

Observe that if a polynomial can be decomposed into a sum of squares (SOS), it is necessarily nonnegative everywhere. Further notice that any sum or product of polynomials nonnegative on a subset is still nonnegative on that subset. Suppose, then, that we have a semialgebraic set $S(\mathbf{g})$ defined as the intersection of the superlevel sets of a tuple $\mathbf{g}$; the corresponding preorder $P(\mathbf{g})$ generated by the g_i is the set of all sums of all combinations of the g_i weighted by SOS polynomials. Schmüdgen proved that polynomials strictly positive on $S(\mathbf{g})$ are members of $P(\mathbf{g})$. The Lasserre hierarchy additionally uses our initial assumption of the Archimedean property of $S(\mathbf{g})$, in which case Putinar's theorem applies, which says that the decomposition of a polynomial nonnegative on an Archimedean $S(\mathbf{g})$ will contain no cross terms of the g_i.

The *quadratic module* $Q(\mathbf{g})$ generated by the tuple $\mathbf{g}$ is exactly the preorder $P(\mathbf{g})$ where all the cross terms weighted by zero:

$$Q(g) = \left\{ s_0 + \sum_{i\in K} g_i s_i \;\middle|\; s_0, s_i \text{ are SOS polynomials } \forall i \in K = 1, \dots, k \right\}. \tag{2.2}$$

Theorem 1 (Putinar's Positivstellensatz [13]). *If a polynomial $p(\mathbf{x})$ is strictly positive for all $\mathbf{x} \in \mathbb{R}^n$ in $S(g_1, \dots, g_k)$, then $p \in Q(g_1, \dots, g_k)$.*

To formulate the sum of squares condition using semidefinite constraints, we define the basis of the linear space of polynomials with degree at most d as

$$[\mathbf{x}]_d = (\mathbf{x}^\alpha)_{\alpha \in \mathbb{N}^m_d}.$$

Any polynomial $p(\mathbf{x})$ with the coefficient vector $\mathbf{c} = (c_\alpha)_{\alpha \in \mathbb{N}^m_d}$ can then be expressed as

$$p(\mathbf{x}) = \sum_{|\alpha| \le d} c_\alpha \mathbf{x}^\alpha = \mathbf{c}^\intercal [\mathbf{x}]_d.$$

A polynomial $p(\mathbf{x})$ of degree at most $2d$ is SOS if, and only if, there exists a positive semidefinite matrix G such that $p(\mathbf{x}) = [\mathbf{x}]_d^\intercal G [\mathbf{x}]_d$ [7, Proposition 2.1]. We can therefore formulate the problem of checking whether such $p(\mathbf{x})$ is SOS as the semidefinite feasibility problem for a matrix G indexed by $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$:

$$\sum_{\beta+\gamma=\alpha} G_{\beta\gamma} = c_\alpha \quad \forall \alpha, |\alpha| \le 2d,$$
$$G \succeq 0.$$

Membership of $p(\mathbf{x})$ in $Q(S)$ can then be formulated as a single semidefinite program by applying such conditions to each polynomial variable in Theorem 1 [7, Chapter 2.4.2].

2.1.3 Hierarchy of Semidefinite Relaxations

The final step is to put bounds on the sizes of moment sequences and degrees of polynomials using the Lasserre's hierarchies [7, Chapter 6]. Each relaxation in the hierarchy is a convex semidefinite program that approximates the solution of the original program.

We consider the quadratic module $Q_h(G)$ truncated to a degree h:

$$Q_h(\mathbf{g}) = \left\{ s_0 + \sum_{i \in K} g_i s_i \;\middle|\; \begin{array}{l} s_0, s_i \text{ SOS polynomials,} \\ \deg(g_i s_i) \le h, \forall i \in K, \\ \deg(s_0) \le h \end{array} \right\}$$

where $\mathbf{g} = \{g_i \mid i \in K = 1, \dots, k\}$. Instead of testing polynomial positivity by checking its membership in $Q(\mathbf{g})$ the hth level of the hierarchy tests for membership in $Q_h(\mathbf{g})$. Similarly, instead of requiring that the constraints hold for all moments, the h-th program in a hierarchy only constrains the moment and localizing matrices of order h. Moreover, we consider only truncated moment sequences $\hat{\mathbf{y}}$, which are just finite-dimensional real vectors. Under our initial Archimedean assumption, the optimization problems are mutually strongly-dual. Each level of the hierarchy is then a semidefinite program which under-estimates the optimum. By going to higher levels, the original problem can be approximated arbitrarily well.

2.1.4 Recovery of Global Minimizers

It is possible to certify the convergence of the Moment hierarchy and to extract the minimizers in a robust way [6]. On the other hand, neither task is directly possible from the side of SOS polynomials, except via duality.

Theorem 2 (Curto and Fialkow [2, Consequence of Theorem 1.6]). *The sequence* $\mathbf{y}$ *of* $2t$ *moments has a representing measure* μ *on a semialgebraic set* $\mathcal{S}(\mathbf{g})$*, if and only if [8]:*

$$\begin{aligned} M_t(\mathbf{y}) &\succeq 0, \\ M_{t-v}(g_i\mathbf{y}) &\succeq 0 \quad \forall g_i \in \mathbf{g}, \\ \operatorname{rank} M_t(\mathbf{y}) &= \operatorname{rank} M_{t-v}(\mathbf{y}), \end{aligned}$$

where $v = \left\lceil \frac{1}{2}\deg(\mathbf{g}) \right\rceil$. *Furthermore* μ *is r-atomic, with* $r = \operatorname{rank} M_d(\mathbf{y})$.

Laurent [10, Section 6.7] explains how software such as *GloptiPoly* and *SumOfSquares* extract solutions from a truncated sequence of moments.   show how to extract, copy it from masters thesis Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

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The convergence rate of the hierarchy is not yet well understood, and it may not be practically possible to reach levels high enough to certify convergence. Regardless of whether a truncated sequence describes a measure or a generic linear functional, we may try the first-order moments as candidate minimizers. The hierarchies yield a nondecreasing sequence of lower bounds on the optimal value of the original problem, while any feasible evaluation of the original problem provides an upper bound. If the first-order moments are feasible and the evaluation of the original problem equals the value of the relaxation, then we have a lower bound equal to the upper bound and therefore a proof of optimality.

In other cases, it may still be possible to glean some insight into the set of minimizers through sublevel sets of the Christoffel-Darboux polynomial [9].

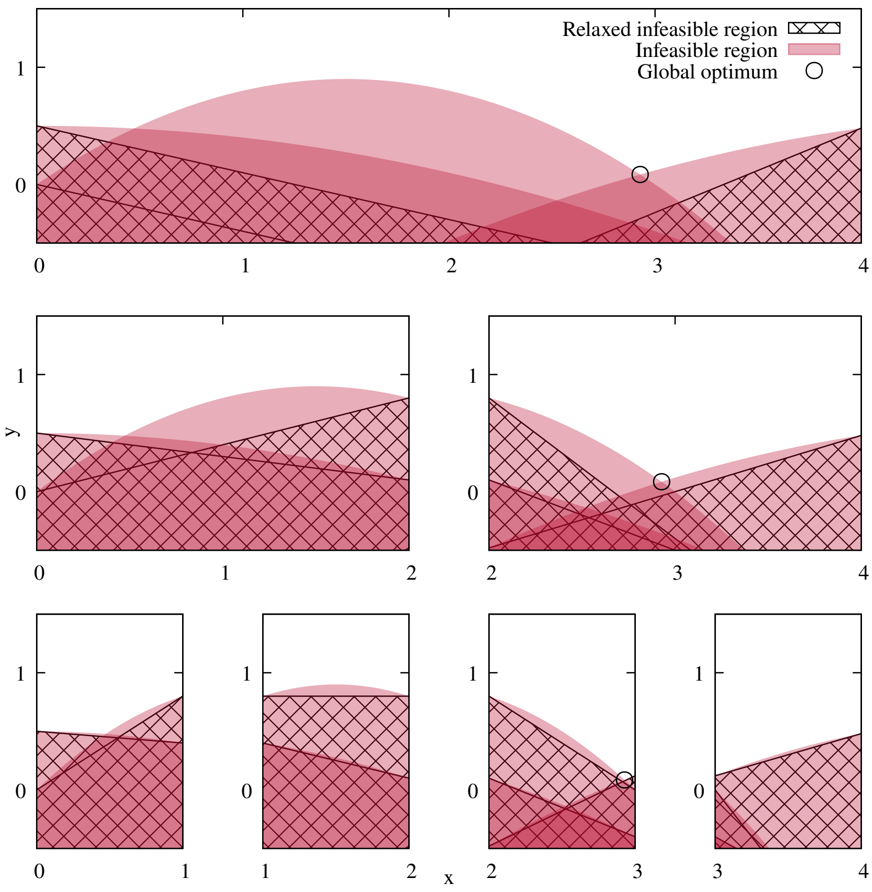

Figure 2.1: An example of a Spatial Branch and Bound tree of a non-convex program minimizing the variable y over $x \in [0, 4]$ such that it lies above the indicated quadratic constraints. The infeasible region is shaded in red, and the locally-valid relaxations via secant cuts are hatched in black.

2.2 Spatial Branch and Bound

Spatial Branch and Bound (BB) is a method to find global optima of non-linear mathematical programs by means of splitting the feasible region in a tree-like pattern and constructing locally-valid cuts from its constraints. BB maintains upper and lower bounds on the global optimum, and the gap between them serves as a proof on how suboptimal the current solution could be. When the gap is smaller than a user-specified threshold, the algorithm stops.

Each step of the algorithm is a branch associated with a set of local bounds on the variables. At each step, the method constructs a convex relaxation for each constraint separately; the relaxed constraints are then solved, e.g. using linear programming. Solutions of the relaxations are lower-bounds on the optimum within the branch and any of its child branches. The worst-case lower-bound in any currently active branch is then a lower bound on the global optimum of the original problem. To improve the lower-bounds, a variable is selected to branch on and its bounds are split into several sub-regions. At the same time, any admissible point for the original problem is an upper-bound on the global optimum. While the program could branch deep enough for the local relaxation to yield an admissible point, it is usual to use heuristics to accelerate the search for feasible points. Any branch where the relaxed sub-problem gives a value worse than an admissible solution can be pruned away. Fig. 2.1 illustrates how BB uses progressively tighter locally-valid cuts to improve the approximation of a concave feasible region defined by quadratic inequalities.

In the rest of the paper, we will show that BB can be performant and scale to large problem sizes, on top of being widely applicable. Even in cases where established techniques exist, such as Lasserre's hierarchies applied to polynomial optimization, BB methods can be competitive. In fact, recent results [5] indicate that considering random instances of some problems results in polynomial time complexity for BB methods.

2.2.1 Convex Relaxations

During the bounding phase of BB, the algorithm constructs a relaxation of the original problem, transforming it into a form that is practically solvable. This likely means relaxing the original problem into a convex one. This process relies on the ability to create reasonable relaxations of the constraints. The literature provides tight convex relaxations for common nonconvex terms, including bilinear and fractional terms, exponentiation, and wholly convex or concave functions. Although tight relaxations may not be generically available, the existing set of relaxations is sufficient to address all algebraic functions [14].

Algebraic functions can be decomposed into a hierarchy of binary operations, each corresponding to one of the aforementioned functions. Consequently, all algebraic functions can be solved using BB by introducing auxiliary variables and appropriate constraints. An implementation of such an algorithm is currently available in the Couenne solver [11]. As an example, to address the multilinear constraint $xyz = 0$, we could introduce a lifting variable $\alpha = xy$ and rewrite the original constraint as $\alpha z = 0$. The result are two constraints for which tight convex relaxations exist.

Symbolic reformulations increase the dimensionality of the problem, which is a significant concern given that the worst-case complexity of the branch-and-bound algorithm is exponential in the number of variables. However, in Chapter **??**, we will demonstrate that these transformations can be competitive when compared to Lasserre's hierarchies, particularly for inverse kinematics of generic redundant manipulators. The polynomial problems arising from inverse kinematics constraints involve more than fourteen variables and degrees greater than four.

2.2.2 Global Optimality Certificates

In comparison to the obvious certificate of global optimality obtained when a feasible solution is found by an outer approximation method such as the Lasserre's hierarchies described in Section 2.1, proving the optimality for the spatial Branch and Bound is less direct. While solvers could return a certificate proving that the solution found is globally optimal within a predefined margin of error, this would reveal the cuts taken by the solver. In the rest of this manuscript, we will focus on the Gurobi solver, which uses a closed-source variant of the spatial BB method and does not return such certificates. The found optimal solution can be used to confirm Gurobi's final primal bound. The log output contains the BB tree information, but it seems impossible to completely reconstruct Gurobi's BB tree or view the specific added cuts. Admittedly, using Gurobi means a certain trade-off between certifying the global optimality and efficiency of the computations. In our experiments, we verified the solutions of BB metods by comparing them with Lasserre's hierarchies where possible.

2.2.3 Finding Feasible Solutions

Getting feasible solutions as early as possible in the BB tree is essential for performance. Using an NLP solver on the problem is an obvious heuristic for finding feasible solutions. However, reformulating problems such that they can be solved by BB methods, e.g. by means of introducing auxiliary variables, can make result in a more complex program and hide some of the original structure. As a result, an NLP solver executed before any reformulation may have an easier time finding a feasible solutions even if the BB solver attempts the same.

Additionally, there may already exist domain-specific methods for finding heuristic solutions with significantly lower complexity than BB methods. For example, in robotics, there exist methods that work in real-time. These methods could be useful both for warm-starting and as runtime heuristics for BB.

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